Courier Management System (CMS)

User Story:

Courier and logistics companies often struggle to efficiently manage shipments and deliveries due to reliance on manual processes and disconnected systems. The absence of a centralized shipment tracking mechanism makes it difficult to monitor parcel movement in real time, resulting in delayed delivery updates and poor visibility for customers. Delivery agent assignments are performed manually, increasing the chances of scheduling conflicts and operational delays. Furthermore, the lack of automated notifications prevents timely communication with customers, while the absence of performance analytics limits managerial insight into operational efficiency. Managing multiple branches becomes complex without an integrated platform, leading to coordination issues across locations. Collectively, these challenges result in customer dissatisfaction, reduced productivity, operational inefficiencies, and ultimately revenue loss for the organization.

1. INTRODUCTION

1.1 Project Overview :

Courier and logistics companies face challenges in efficiently managing shipments and deliveries due to manual operations and disconnected systems. These issues lead to delayed updates, poor customer visibility, and operational inefficiencies.

The Courier Management System (CMS) is a Salesforce-based cloud solution that centralizes shipment booking, tracking, delivery agent assignment, automated status updates, billing, and analytics. The system improves operational efficiency, enhances customer satisfaction, and provides real-time business insights.

1.2 Purpose :

The purpose of the Courier Management System (CMS) is to automate and centralize courier and logistics operations using the Salesforce platform. The system aims to replace manual and fragmented processes with an integrated digital solution that enables efficient shipment booking, real-time tracking, delivery agent assignment, automated notifications, billing, and performance reporting. By providing a single, unified platform for managing shipments across multiple branches, CMS improves operational visibility, enhances customer communication, reduces delays and errors, and supports faster decision-making through analytics. Ultimately, the system is designed to increase service efficiency, customer satisfaction, and overall business profitability.

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	**I am (Customer)**	**I'm trying to**	**But**	**Because**	**Which makes me feel**
PS-1	Branch Manager	manage shipments efficiently	tracking is manual	systems are disconnecte d	stressed
PS-2	Customer	track parcels	updates are delayed	no centralized system	uncertain

2.2 Empathy Map Canvas

User Persona: Branch Manager

- **Think & Feel:** Concerned about delays & errors
- **Hear:** Customer complaints
- **See:** Registers & spreadsheets
- **Say & Do:** Manual follow-ups
- **Pain:** Delivery conflicts & missed updates
- **Gain:** Automated tracking & dashboards

2.3 Brainstorming & Idea Prioritization

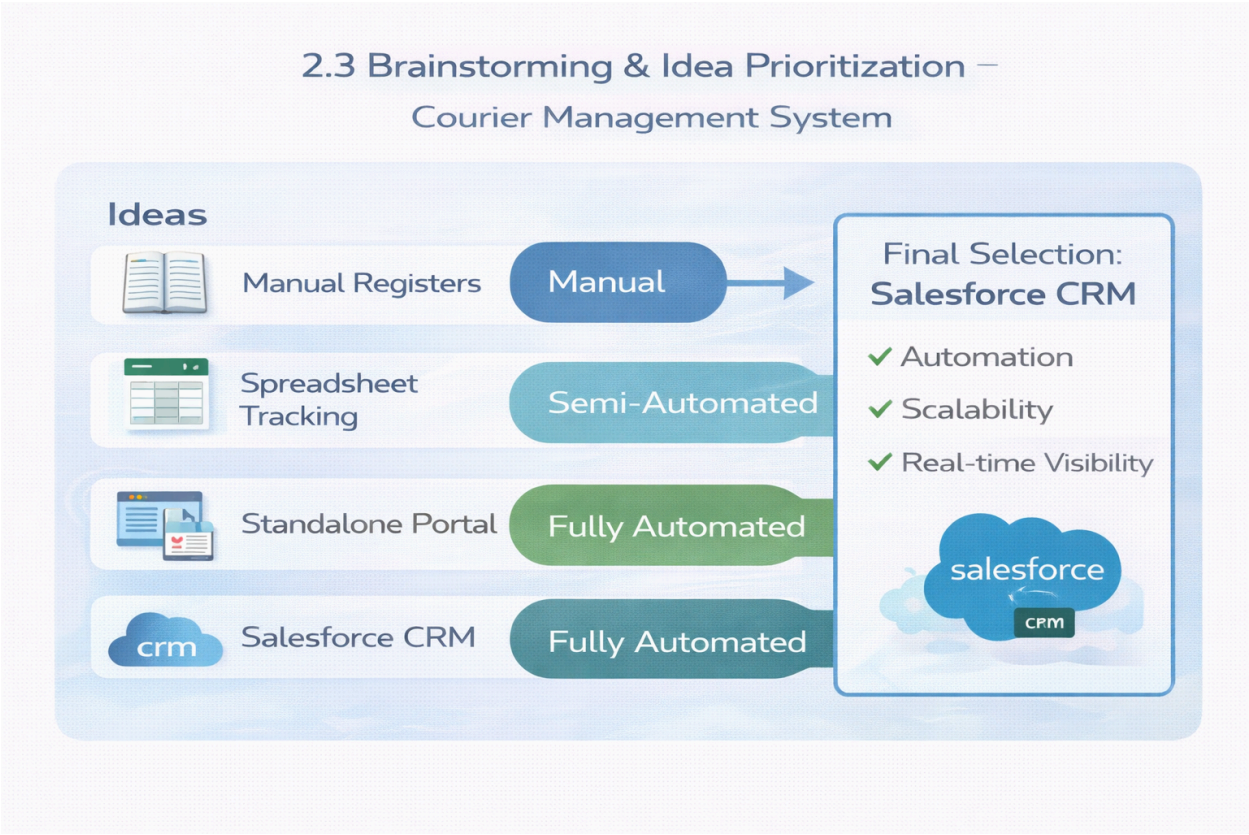
Ideas Considered

- Manual registers
- Spreadsheet tracking
- Standalone portal
- Salesforce CRM

Final Selection

Salesforce CRM selected due to:

- Automation
- Scalability
- Real-time visibility

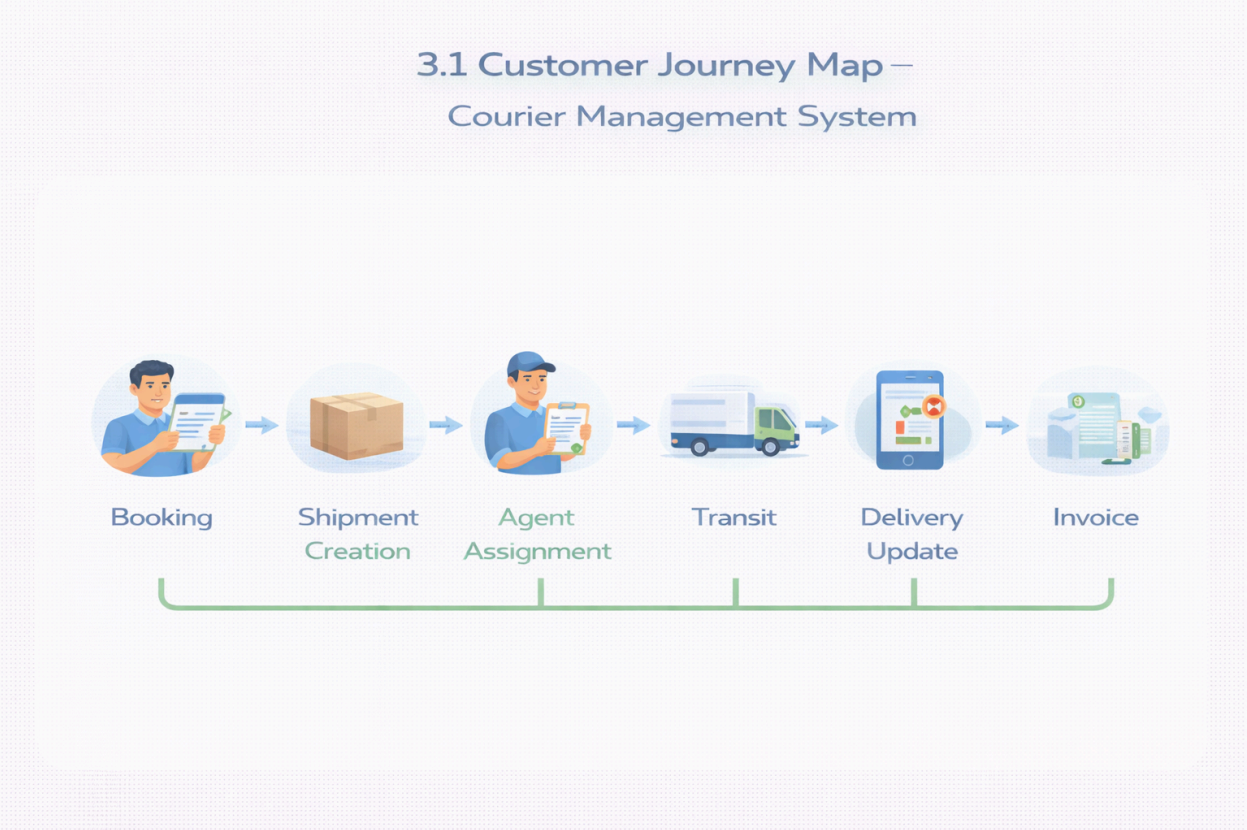


3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages:

- Booking → Shipment Creation → Agent Assignment → Transit → Delivery Update → Invoice → Reporting



3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	Customer management
FR-2	Shipment booking
FR-3	Delivery agent assignment
FR-4	Real-time status updates
FR-5	Invoice generation

FR-6	Reports & dashboards

Non-Functional Requirements

NFR	**Description**
Usability	Easy Lightning interface
Security	Role-based access control
Reliability	Error-free automation
Performance	Fast tracking updates
Availability	24/7 cloud access
Scalability	Supports growing branches

3.3 Data Flow Diagram

Flow:

Customer Input → Shipment Creation → Validation → Agent Assignment → Delivery Update → Invoice → Reports

3.4 Technology Stack

Layer	**Technology**

UI	Salesforce Lightning
Logic	Salesforce Flows
Database	Salesforce Objects
Automation	Validation Rules, Flows
Security	Profiles, Roles
Reporting	Reports & Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

CMS eliminates fragmented courier operations by automating shipment tracking, delivery assignment, billing, and reporting within a centralized Salesforce platform.

4.2 Proposed Solution

Functional Requirements

Parameter	**Description**
Problem	Manual shipment management
Solution	Salesforce CMS
Innovation	Automated tracking & billing

Impact	Faster deliveries
Business Model	Courier services
Scalability	Multi-branch ready

4.3 Solution Architecture

Components:

- Customer Object
- Shipment Object
- Delivery Agent Object
- Branch Object
- Delivery Update Object
- Invoice Object
- Flows
- Reports
- Dashboards

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Agile methodology
- Epics → Stories → Story Points
- Sprint-based execution
- Velocity calculation used for estimation

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create and configure a Salesforce developer account so that I can set up and deploy the Courier Management System.	3	High	Member1
Sprint-2	Data Modeling	USN-2	As an admin, I want to create Customer, Shipment, Delivery Agent, Branch, and Invoice objects with proper relationships so that courier data is stored and managed efficiently.	5	High	Member2
Sprint-2	Data Modeling	USN-3	As a staff member, I want tabs and a Lightning App for all courier modules so that I can easily navigate and perform daily operations.	5	High	Member3
Sprint-3	Automation	USN-4	As a system, I want to validate shipment details such as delivery dates, addresses, and mandatory fields so that incorrect or incomplete records are prevented.	3	High	Member4
Sprint-3	Automation	USN-5	As a dispatcher, I want automated flows for shipment status updates and agent assignment so that deliveries are processed faster without manual intervention.	3	High	Member4
print-4	Security	USN-6	As an administrator, I want role-based access control for managers, agents, and staff so that sensitive shipment and billing data is secure.	5	High	Member2
Sprint-5	Repots	USN-7	As a branch manager, I want reports showing delivery performance, shipment status, and revenue so that I can monitor operations and make informed decisions.	4	High	Member3
Sprint-6	Dashboards	USN-8	As a branch manager, I want dashboards showing delivery performance, shipment status, and revenue so that I can monitor operations and make informed decisions.	4	Medium	Member4

Project Tracker, Velocity & Burndown Chart: (4 Marks)

6. Project Development Phase:

What you'll learn

1. Real-time Salesforce project implementation
2. Designing custom objects and relationships
3. Creating page layouts & record types
4. Writing validation rules
5. Building automation with flows
6. Developing reports & dashboards
7. Setting up security with profiles, permission sets, and sharing
8. End-to-end application testing

Milestone 1: Salesforce Account

Introduction:

New to Salesforce and unsure where to start? Curious how real business applications—like loan processing and payment tracking—can be built on the platform? You're in the right place.

Welcome to the Loan Management System project!

Salesforce offers powerful tools that help automate processes, improve accuracy, and manage customer operations with ease. In this project, you'll learn how Salesforce can streamline lending workflows—from borrower onboarding to loan approval, EMI tracking, and document verification.

What Is Salesforce ?

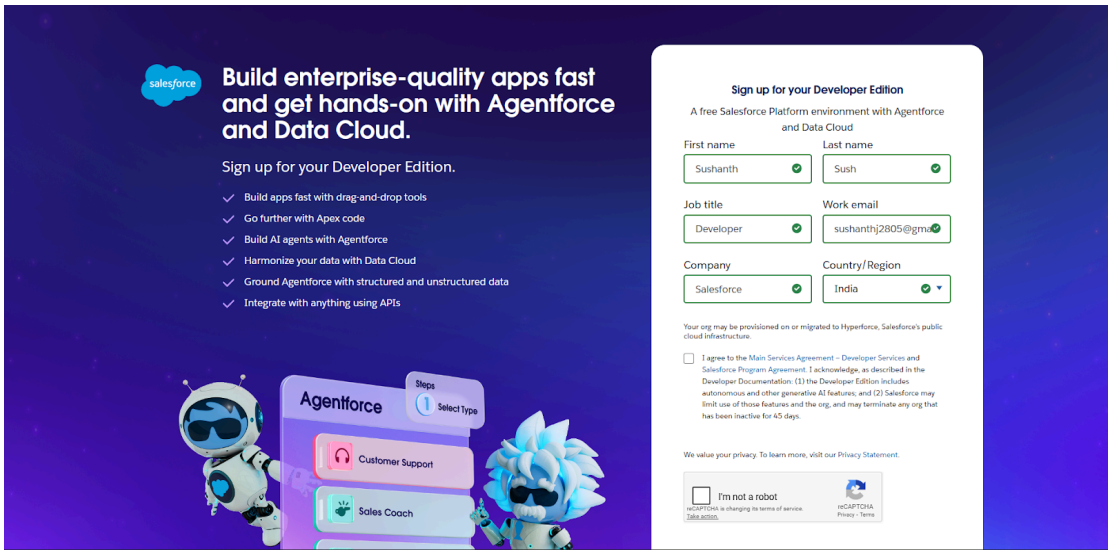
Salesforce is your customer success platform, designed to help you sell, service, market, analyze, and connect with your customers. Salesforce has everything you need to run your business from anywhere. Using standard products and features, you can manage relationships

with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud.

Activity 1: Creating Developer Account

Creating a developer org in salesforce.

1. Go to https://developer.salesforce.com/signup
2. On the sign up form, enter the following details :
3. First name & Last name
4. Email

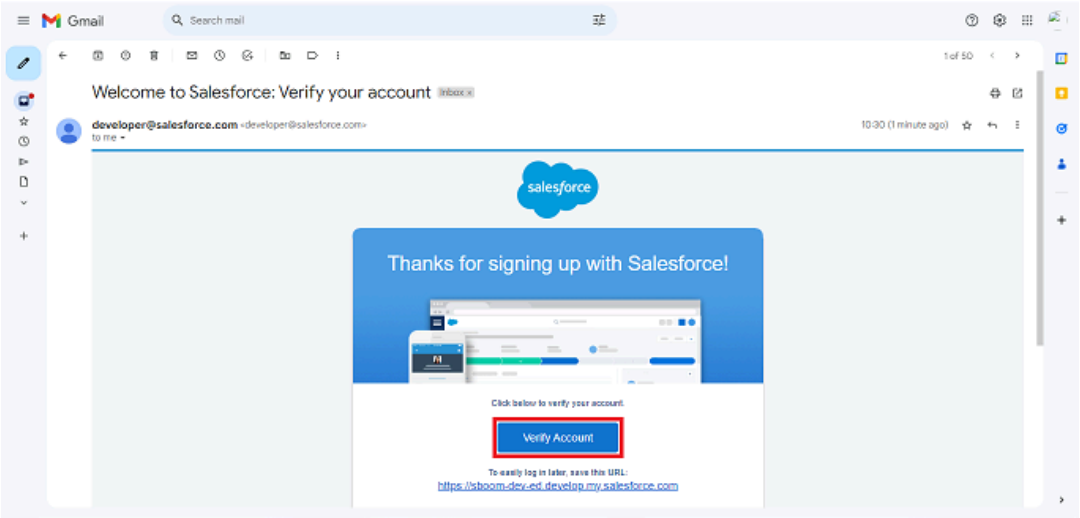


1) First name & Last name
2) Email
3) Job Title: Developer
4) Company: College Name
5) Country: India

Click on sign me up after filling these.

Activity 2: Account Activation

1. Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 10-30 mins and sometimes 2 hours.



2. Click on Verify Account
3.Give a password and answer a security question and click on change password.
4.Then you will redirect to your salesforce setup page.

Milestone 2- Object:

What is an object?

Salesforce objects are database tables that permit you to store data that is specific to an organization. It consists of fields (columns) and records (rows).

Custom Objects Created:

1. **Customer__c** – stores customer personal and contact information

2. **Shipment__c** – manages parcel details, source, destination, and tracking information

3. **Delivery_Agent__c** – maintains delivery agent profiles and branch details

4. **Branch__c** – represents courier branch offices

5. **Delivery_Update__c** – stores status updates for each shipment

6. **Invoice__c** – manages billing details for delivered shipments

Relationships:

Customer → Shipment (Lookup)

Shipment → Delivery_Agent (Lookup)

Delivery_Agent → Branch (Lookup)

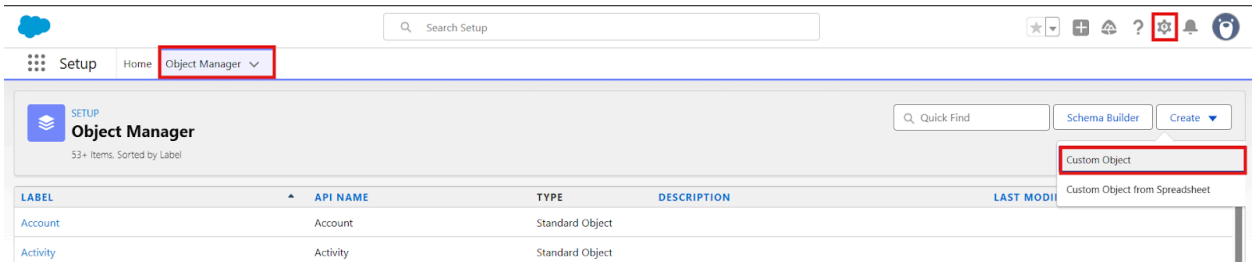
Delivery_Update → Shipment (Master-Detail)

Invoice → Shipment (Lookup)

Steps to Implement :

Activity 1: Create Customer Object

1. Click Setup → Object Manager → Create → Custom Object



2. Enter the following details:

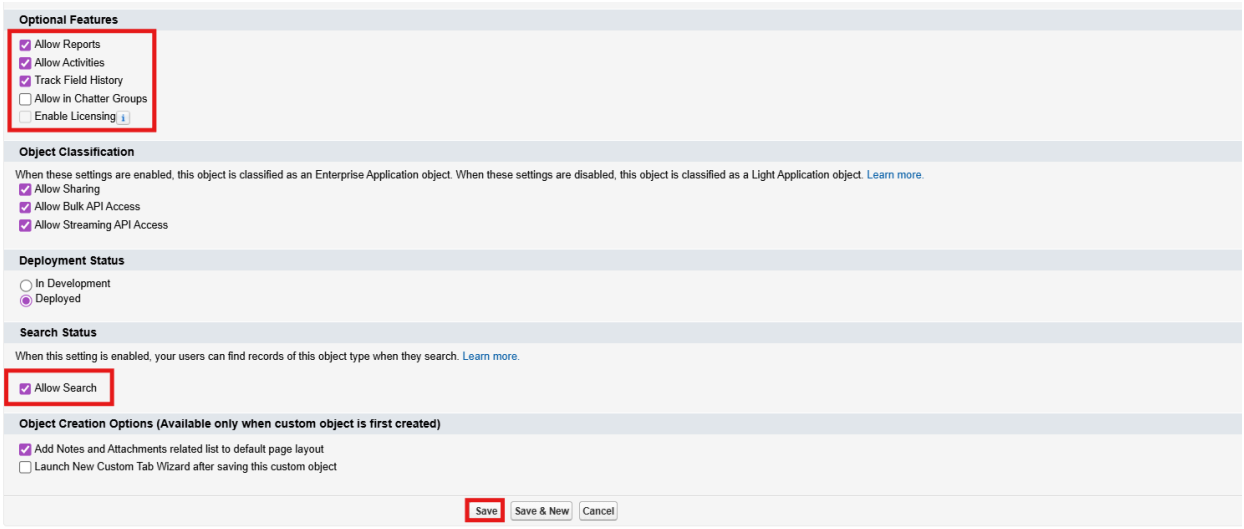
- Label: Customer
- Plural Label: Customers

- Record Name: Customer Name (Text)

3. Under Optional Features:

- Enable Allow Reports
- Enable Track Field History
- In Search Status check the box Allow Search
- In Object Creation Options check the box Add Notes and Attachments related list to default page layout

4. Click Save.



Activity 2: Create Shipment Object

1. Go to Setup → Object Manager → Create → Custom Object

2. Enter:

- Label: Shipment
- Plural Label: Shipments
- Record Name: Tracking Number (Auto Number)

3. Example Format: SHP-{00000}
   Starting Number: 0
4. Enable:

- Allow Reports
- Track Field History

5. Click Save

Activity 3: Create Delivery_Agent Object

1. Setup → Object Manager → Create → Custom Object
2. Enter:

- Label: Delivery Agent
- Plural Label: Delivery Agents
- Record Name: Agent Name (Text)

3. Enable:

- Allow Reports
- Track Field History

4. Click Save

Activity 4: Create Branch Object

1. Setup → Object Manager → Create → Custom Object
2. Enter:

- Label: Branch
- Plural Label: Branches
- Record Name: Branch Name (Text)

3. Enable:
- Allow Reports

4. Click Save.

Activity 5: Create Delivery_Update Object

1. Setup → Object Manager → Create → Custom Object
2. Enter:
- Label: Delivery Update
- Plural Label: Delivery Updates
- Record Name: Update Number (Auto Number)

3. Example Format: UPD-{0000}
   Enter Starting number : 0
4. Enable:
- Allow Reports
- Track Field History

5. Click Save.

Activity 6: Create Invoice Object

1. Setup → Object Manager → Create → Custom Object
2. Enter:

- Label: Invoice
- Plural Label: Invoices
- Record Name: Invoice Number (Auto Number)

3. Example Format: INV-{0000}
   Enter Starting number: 0
4. Enable:

- Allow Reports
- Track Field History

5. Click Save

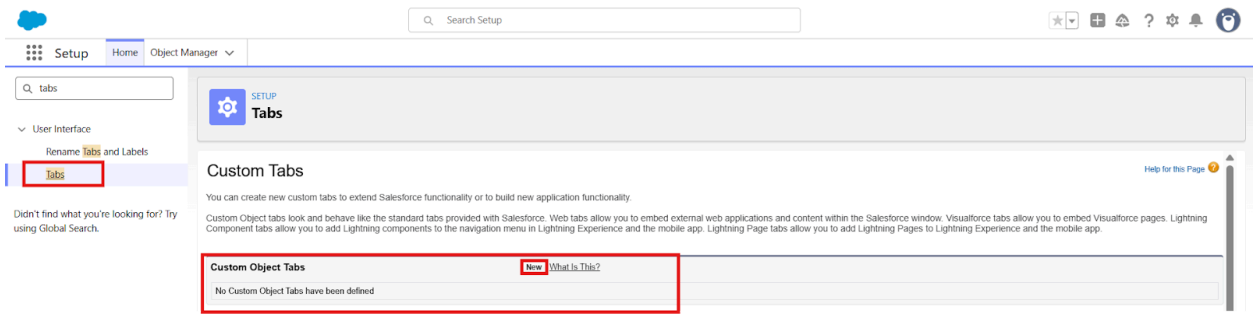
Milestone 3 - Tabs

Tab:

Tabs in Salesforce help users view the information at a glance. It displays the data of objects and other web content in the application.

Activity 1: Creating a Tab for Customer Object

1. Click Home, type Tabs in Quick Find, and select Tabs.
2. Under Custom Object Tabs, click New.



3. For Object, select Customer.
4. For Tab Style, choose any icon.
5. Leave all default settings as they are.
6. Click Next, Next, and Save.

Activity 2: Creating a Tab for Shipment Object

1. Click Home, type Tabs in Quick Find, and select Tabs.
2. Under Custom Object Tabs, click New.
3. For Object, select Shipment.
4. Choose any Tab Style.
5. Keep defaults unchanged.
6. Click Next, Next, and Save.

Activity 3: Creating a Tab for Delivery Agent Object

1. Click Home, type Tabs in Quick Find, and select Tabs.
2. Under Custom Object Tabs, click New.
3. For Object, select Delivery Agent.

4. Pick any Tab Style.
5. Leave default settings.
6. Click Next, Next, and Save.

Activity 4: Creating a Tab for Branch Object

1. Click Home, type Tabs in Quick Find, and select Tabs.
2. Under Custom Object Tabs, click New.
3. For Object, select Branch.
4. Choose any Tab Style.
5. Keep all defaults.
6. Click Next, Next, and Save.

Activity 5: Creating a Tab for Delivery Update Object

1. Click Home, type Tabs in Quick Find, and select Tabs.
2. Under Custom Object Tabs, click New.
3. For Object, select Delivery Update.
4. Select any Tab Style.
5. Keep default configurations.
6. Click Next, Next, and Save.

- Check the Required box
- Click Next → Next → Save

2. Phone (Phone Field)

- Setup → Object Manager → Customer__c
- Fields & Relationships → New
- Select Phone → Next
- Field Label: Phone
- Check Required
- Click Next → Next → Save

3. Address (Text Area)

- Setup → Object Manager → Customer__c
- Fields & Relationships → New
- Select Textarea → Next
- Field Label: Address
- Check Required
- Save

4. Customer Type (Picklist)

- Setup → Customer__c → Fields → New
- Picklist → Next
- Field Label: Customer Type
- Enter values:
    - Regular

    - Business
- Check Required
- Save

5. City (Text)

- Setup → Customer__c → Fields → New
- Text → Next
- Field Label: **City**
- Save

6. State (Text)

- Setup → Customer__c → Fields → New
- Text → Next
- Field Label: **State**
- Save

7. Pincode (Text)

- Setup → Customer__c → Fields → New
- Text → Next
- Field Label: **Pincode**
- Save

8. Is Active (Checkbox)

- Setup → Customer__c → Fields → New
- Checkbox → Next

- Field Label: **Is Active**
- Save

1. Source Address (Text Area)

- Setup → Shipment__c → Fields → New
- Textarea → Next
- Field Label: **Source Address**
- Required → Save

2. Destination Address (Text Area)

- Setup → Shipment__c → Fields → New
- Textarea → Next
- Field Label: **Destination Address**
- Required → Save

3. Weight (Number)

- Setup → Shipment__c → Fields → New
- Number → Next
- Field Label: **Weight**

- Required → Save

4. Price (Currency)

- Setup → Shipment__c → Fields → New
- Currency → Next
- Field Label: **Price**
- Required → Save

5. Status (Picklist)

Values:

- Booked
- In Transit
- Out for Delivery
- Delivered
- Cancelled

Required → Save

6. Expected Date (Date)

- Setup → Shipment__c → Fields → New
- Date → Next
- Field Label: **Expected Date**

- Required → Save

7. Shipment Type (Picklist)

Values:

- Domestic
- International

Required → Save

Activity 3: Creation of fields for Delivery_Agent Object

Field Name	API Name	Data Type	Required
Agent Name	Name	Text	Yes
Email	Email__c	Email	Yes
Phone	Phone__c	Phone	Yes
Branch	Branch__c	Lookup	Yes
Status	Status__c	Picklist	Yes
Joining Date	Joining_Date__c	Date	No

1. Email

- Setup → Delivery_Agent__c → Fields → New
- Email → Next
- Field Label: Email

- Required → Save

2. Phone

- Setup → Delivery_Agent__c → Fields → New
- Phone → Next
- Field Label: Phone
- Required → Save

3. Status (Picklist)

Values:

- Active
- Inactive
- On Leave

Required → Save

4. Joining Date

- Setup → Delivery_Agent__c → Fields → New
- Date → Next
- Field Label: Joining Date
- Save

Activity 4: Creation of fields for Branch Object

Field Name	API Name	Data Type	Required

Branch Name	Name	Text	Yes
Branch Code	Branch_Code__c	Text	Yes
City	City__c	Text	Yes
State	State__c	Text	Yes
Manager Name	Manager_Name__c	Text	No

Create each field using:

- Setup → Branch__c → Fields → New
- Select respective Data Type
- Enter Field Label
- Save

(Repeat for all fields)

Activity 5: Creation of fields for Delivery_Update Object

Field Name	API Name	Data Type	Required
Update Number	Name	Auto Number	Yes
Shipment	Shipment__c	Master-Detail	Yes
Status	Status__c	Picklist	Yes
Update Date	Update_Date__c	DateTime	Yes
Remarks	Remarks__c	Text	No

Status Picklist Values:

- In Transit
- Out for Delivery
- Delivered
- Delayed

Required → Save

Activity 6: Creation of fields for Invoice Object

Field Name	API Name	Data Type	Required
Invoice Number	Name	Auto Number	Yes
Shipment	Shipment__c	Lookup	Yes
Amount	Amount__c	Currency	Yes
Invoice Date	Invoice_Date__c	Date	Yes
Payment Status	Payment_Status__c	Picklist	Yes

Payment Status Values

- Paid
- Unpaid
- Pending

Required → Save

Relationships :

Child Object	Field	Parent Object	Type
Shipment	Customer__c	Customer	Lookup
Shipment	Assigned_Agent__c	Delivery_Agent	Lookup
Delivery_Agent	Branch__c	Branch	Lookup
Delivery_Update	Shipment__c	Shipment	Master-Detail
Invoice	Shipment__c	Shipment	Lookup

Relationship Creation Steps :

Activity 1 – Shipment → Customer

- Setup → Shipment__c → Fields → New
- Lookup Relationship
- Related Object: **Customer__c**
- Field Label: **Customer**
- Save

Activity 2 – Shipment → Delivery Agent

- Setup → Shipment__c
- New → Lookup
- Object: **Delivery_Agent__c**

- Label: **Assigned Agent**
- Save

Activity 3 – Delivery Agent → Branch

- Setup → Delivery_Agent__c
- New → Lookup
- Related Object: **Branch__c**
- Label: **Branch**
- Save

Activity 4 – Delivery Update → Shipment

- Setup → Delivery_Update__c
- New → **Master-Detail**
- Related Object: **Shipment__c**
- Save

Activity 5 – Invoice → Shipment

- Setup → Invoice__c
- New → Lookup
- Object: **Shipment__c**
- Save

Milestone 5-Validation Rules :

Validation rules are applied when a user tries to save a record and are used to check if the data meets specified criteria. If the criteria are not met, the validation rule triggers an error message and prevents the user from saving the record until the issues are resolved.

Activity 1: Create Validation Rules on Customer__c

1. Validation Rule: Email Must Not Be Blank

Steps:

- Click the Gear icon → Setup
- Go to Object Manager
- Search and select Customer__c
- From the left menu, click Validation Rules
- Click New
- Rule Name: Email_Required
- Check Active

In Error Condition Formula, paste:

ISBLANK(Email__c)

- Click Check Syntax
- Error Message: Email is required.

- Error Location: Field → Email
- Click Save.

2. Validation Rule: Phone Must Be 10 Digits

Steps:

- Setup → Object Manager → Customer__c
- Validation Rules → New
- Rule Name: Validate_Phone_Number
- Check Active

Formula:

LEN(Phone__c) <> 10

- Check Syntax
- Error Message: **Phone number must be 10 digits.**
- Error Location: Field → Phone
- Save

3. Validation Rule: Pincode Must Be 6 Digits

Steps:

- Setup → Object Manager → Customer__c
- Validation Rules → New
- Rule Name: Validate_Pincode

- Active → Yes

Formula:

LEN(Pincode__c) <> 6

- Error Message: **Pincode must be 6 digits.**
- Location: Field → Pincode
- Save

Activity 2: Create Validation Rules on Shipment__c

1. Validation Rule: Weight Must Be Greater Than Zero

Steps:

- Setup → Object Manager → Shipment__c
- Click Validation Rules → New
- Rule Name: Validate_Weight
- Active → Checked

Formula:

Weight__c <= 0

- Check Syntax
- Error Message: **Weight must be greater than zero.**
- Location: Field → Weight
- Save

2. Validation Rule: Price Must Be Greater Than Zero

Steps:

- Setup → Object Manager → Shipment__c
- Validation Rules → New
- Rule Name: Validate_Price
- Active → Yes

Formula:

Price__c <= 0

- Error Message: **Price must be greater than zero.**
- Location: Field → Price
- Save

3. Validation Rule: Expected Date Cannot Be in the Past

Steps:

- Setup → Object Manager → Shipment__c
- Validation Rules → New
- Rule Name: Validate_Expected_Date

Formula:

Expected_Date__c < TODAY()

- Check Syntax
- Error: **Expected delivery date cannot be in the past.**

- Location: Field → Expected Date
- Save

7. FUNCTIONAL & PERFORMANCE TESTING

During testing, screenshots were captured from Salesforce to verify record creation, validation rule enforcement, delivery status updates, flow automation, invoice generation, and dashboard visualization. All scenarios were validated to ensure correct performance and reliability.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data
- Automated workflows
- Real-time reporting

Disadvantages

- Salesforce dependency
- Requires initial configuration effort

9. Conclusion:

The Courier Management System successfully digitizes courier operations using Salesforce automation, improving efficiency, accuracy, and customer satisfaction.

10. FUTURE SCOPE

- Mobile tracking app
- GPS integration
- SMS notifications
- AI-based delivery optimization

11. APPENDIX

- Source Code: Salesforce Configuration
- Dataset: Salesforce Objects
- GitHub / Demo: Salesforce Developer Org

Thank You